

A Low-Cost Helmet-Integrated System for Measuring Head Impacts in American Football Players

Juan Ramón Lara Mora Montserrat Sánchez Chávez Mariel Alfaro Ponce
Gerardo Julián Ramírez-Nava

Tecnológico de Monterrey, Campus Ciudad de México, School of Engineering and Sciences

Abstract

Background: Repetitive head impacts in American football can lead to suspected sport-related concussion and long-term neurological risk. Commercial helmet telemetry systems can support screening, but their cost and importation barriers can limit adoption in resource-constrained collegiate and youth programs. **Objective:** This work presents the design and preliminary benchtop validation of a low-cost helmet-integrated system that measures linear acceleration and estimates impact location in an American football helmet. **Methods:** A Schutt F7 VTD Collegiate helmet was instrumented with an H3LIS200DL triaxial accelerometer, four SEN-10245 load cells with HX711 conditioning modules, and a Raspberry Pi Zero 2 W controller. The embedded algorithm triggers impact recording above 14.4 g, samples the sensor response for four seconds at 40 ms intervals, estimates the dominant impact region from the maximum load-cell response, and displays an alert for coaching staff. Benchtop validation used a pendulum-based accelerometer comparison against a commercial BWT61CL unit and static load-cell calibration with known masses. **Results:** The prototype achieved approximately 94 % agreement with the commercial accelerometer in repeated pendulum trials and a reported 86 % load-cell accuracy in static calibration. The direct hardware cost was approximately US\$95.90 for one helmet prototype, below the reported cost of commercial impact-monitoring alternatives. **Conclusion:** The proposed system is a feasible proof-of-concept for low-cost head-impact monitoring and sideline alerting. Before field deployment, the design requires dynamic impact validation, rotational kinematics, water and temperature qualification, multi-user data management, and clinical workflow integration. The device should be treated as a screening adjunct and not as a diagnostic or return-to-play decision tool.

Keywords: sport-related concussion; American football; helmet instrumentation; accelerometer; load cell; head-impact biomechanics; low-cost medical device prototype

1 Introduction

American football exposes athletes to repeated head acceleration events that vary by position, play type, and impact location. Although helmet technology has improved substantially, the helmet alone does not eliminate the need for impact surveillance, sideline screening, and conservative clinical management after a suspected concussion. Current sport-concussion guidance emphasizes multimodal assessment and medically supervised return-to-sport decisions rather than reliance on a single sign, symptom, or sensor value [1, 2]. Repetitive head impacts and recurrent concussions have also been associated with later-life cognitive and mood outcomes in retired players [3–5].

Instrumented helmets and mouthguards have enabled quantitative research on head-impact exposure. Prior studies in collegiate and high-school football have shown that head-impact magnitude

and frequency differ by position and event context [6–8]. These systems, however, are not always accessible to student teams because of cost, importation logistics, proprietary software, and the need for trained personnel to interpret reports. The gap is particularly relevant for teams that rely on internal protocols and may lack routine access to dedicated sports-medicine personnel at every training session.

The present study addresses this gap by converting a course-project prototype into a concise research article. The objective is to describe a low-cost, helmet-integrated system that combines an accelerometer and four load cells to (i) detect head-impact events above a selected acceleration threshold, (ii) estimate the region of impact, and (iii) provide a simple sideline alert. The system is intended to support earlier recognition of potentially hazardous impacts and to complement, not replace, established concussion protocols.

2 Design requirements and use context

The design was developed for the FBA program at Tecnológico de Monterrey, Campus Ciudad de México, where coaches identified the need for faster recognition of impacts that may otherwise be underreported by athletes. The intended users are coaches and medical staff who require simple visual feedback during training or controlled testing. The device is not intended to diagnose concussion, clear an athlete for return to play, or substitute for medical evaluation.

Table 1 summarizes the translated engineering requirements used to guide the prototype. The requirements prioritize measurement and alerting while keeping the system low-cost, compact, and compatible with the internal geometry of a football helmet.

Table 1: Engineering requirements for the helmet-integrated impact-monitoring prototype.

Category	Requirement
Must have	Measure linear acceleration and load-cell response during head-impact events; identify the dominant impact region; avoid interfering with helmet fit as much as possible.
Should have	Apply configurable impact thresholds; display results on an external screen; store event records; provide clear alert levels for coaching staff.
Could have	Wireless data export, longer battery autonomy, multi-athlete profiles, and a field-ready enclosure.
Out of scope	Medical diagnosis, automatic return-to-play decisions, and replacement of established concussion-screening tools.

3 Methods

3.1 Biomechanical model

The system uses linear acceleration as the primary impact trigger because peak linear acceleration is widely used in helmet-impact surveillance and is directly measurable with compact inertial sensors [6, 9]. The relationship between impact force and acceleration is approximated from Newton’s second law,

$$F = ma, \tag{1}$$

where F is force, m is the effective mass participating in the event, and a is linear acceleration. Rotational dynamics were recognized as clinically relevant but were not implemented in this prototype. In a future version, angular acceleration could be related to torque by

$$\tau = I\alpha, \quad (2)$$

where τ is torque, I is moment of inertia, and α is angular acceleration.

Momentum conservation was used as a conceptual model for player-player impacts,

$$m_1v_1 + m_2v_2 = m_1v_3 + m_2v_4, \quad (3)$$

where m_1 and m_2 represent the effective masses of the impacted and striking players, respectively, and v_1 – v_4 represent pre- and post-impact velocities. This approximation highlights the role of neck and torso alignment in changing the effective mass during contact [10, 11].

The load-cell subsystem estimates the dominant impact region from strain-induced resistance changes. Strain is defined as

$$\varepsilon = \frac{\Delta L}{L_0}, \quad (4)$$

and the relative resistance change of a strain gauge is

$$\frac{\Delta R}{R} = K\varepsilon, \quad (5)$$

where K is the gauge factor [12]. Each load cell was conditioned through an HX711 module, which provides amplification and analog-to-digital conversion for bridge-based load sensing.

3.2 Hardware architecture

The prototype consists of three integrated subsystems: sensing, processing, and reporting. The sensing layer combines one triaxial accelerometer and four load cells. The processing layer is a Raspberry Pi Zero 2 W. The reporting layer displays the maximum acceleration and estimated impact location on an external laptop interface.

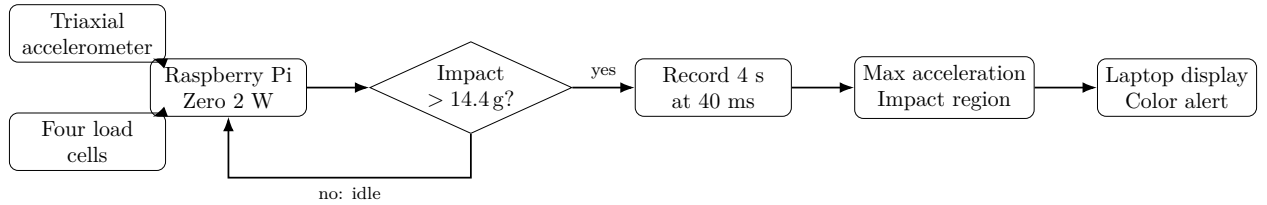


Figure 1: Functional architecture of the proposed head-impact monitoring system. The acceleration threshold triggers event recording, after which the system computes peak acceleration and the load-cell channel with the largest response.

Table 2 lists the main electronic components. The accelerometer supports selectable high- g ranges and digital communication through I²C or SPI [13]. The Raspberry Pi Zero 2 W was selected because it provides wireless connectivity, a compact form factor, GPIO access, and sufficient processing for the prototype interface [14]. Four load cells were placed to represent frontal, posterior, and lateral regions of the helmet.

Table 2: Main hardware components and prototype role.

Component	Model	Role
Single-board computer	Raspberry Pi Zero 2 W	Controls data acquisition, threshold logic, storage, and external display.
Accelerometer	H3LIS200DL / SEN0408	Measures linear acceleration in high- g impact events.
Load cells	SEN-10245, 50 kg each	Estimate impact region from local load response.
Signal conditioner	HX711 modules	Amplifies and digitizes load-cell bridge output.
Battery module	2200 mAh rechargeable pack	Supports untethered prototype operation; autonomy requires field validation.

3.3 Helmet integration

A Schutt FB 208800 F7 VTD Collegiate football helmet was used as the integration platform. The design places the accelerometer near the superior-anterior helmet region and distributes the four load cells near the regions expected to receive frontal, rear, and lateral contacts. The electronics were positioned between the helmet shell and padding system to minimize interference with athlete comfort.

3.4 Embedded algorithm

The acquisition program was written in Python. During idle operation, the system monitors acceleration and load-cell channels. When the acceleration exceeds 14.4 g, the system records approximately four seconds of event data at 40 ms intervals. For each event, the software computes the maximum acceleration observed and compares the maximum values from the four load-cell channels. The channel with the highest response is reported as the dominant impact region. A color-coded output was implemented to simplify interpretation by non-engineering users.

The 14.4 g threshold was selected as an initial activation point inspired by commercial impact-alert systems and the need to reduce false logging during normal handling. A higher warning threshold of 50 g was included for severe events in the prototype interface. These values should not be interpreted as diagnostic concussion thresholds, because concussion risk depends on linear and rotational kinematics, impact duration, prior injury history, symptoms, age, sex, and clinical context [1, 2].

3.5 Calibration and validation procedures

Accelerometer validation used a pendulum setup with a 13 cm string and a release angle of approximately 60°. The tangential component of gravitational acceleration was used as a theoretical reference,

$$F_t = -mg \sin \theta = ma_t, \quad (6)$$

where θ is the pendulum angle and a_t is tangential acceleration. The instrumented H3LIS200DL measurements were compared with a commercial WitMotion BWT61CL accelerometer across ten repetitions.

Load-cell validation used a flat board to distribute known masses across the sensor arrangement. The calibration sequence included a 190 g reference object and nominal 2.5 lb, 5 lb, and 10 lb test



(a) Front view.



(b) Top view.

Figure 2: CAD-based sensor placement used to guide helmet instrumentation. The extracted project figures show the proposed placement of the accelerometer and load-cell positions relative to the helmet shell and padding.

masses. The load-cell system was tared before measurement. Because the prototype used static vertical loading rather than helmet-like dynamic impacts, these tests validate only preliminary calibration and not final field accuracy.

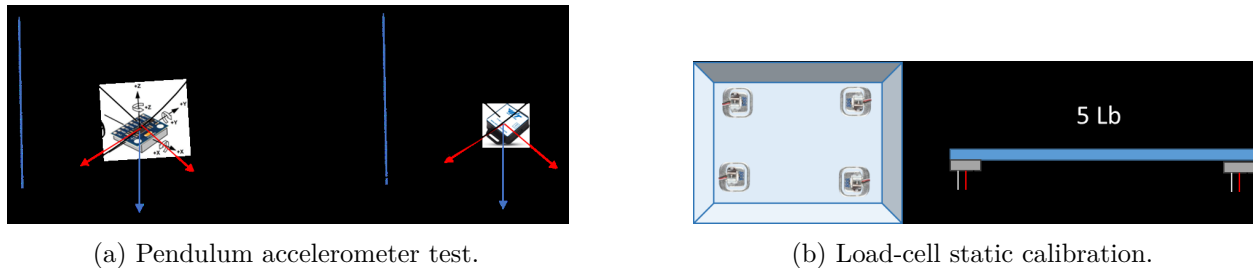


Figure 3: Benchtop validation setups. The pendulum test provided a repeatable acceleration input, while the load-cell test used known masses to evaluate the static response of the force-sensing layer.

4 Results

4.1 Prototype implementation

The prototype was assembled using a fabric insert shaped to follow the internal helmet padding. Load cells were attached to the fabric layer and routed to HX711 modules connected to the Raspberry Pi. The approach allowed the sensing layer to be placed under the helmet padding while preserving the original shell structure. Figure 4 shows the assembled insert before final closure.

The direct hardware cost of the prototype was approximately US\$95.90 (Table 3). This cost does not include labor, software development, testing time, importation fees, or casing for a field-ready product. The original project estimated approximately 154 hours of development work.

Table 3: Approximate direct hardware cost for one prototype.

Item	Quantity	Unit cost (US\$)	Total (US\$)
H3LIS200DL / SEN0408 accelerometer	1	14.90	14.90
50 kg SEN-10245 load cell with HX711 module	4	5.00	20.00
Raspberry Pi Zero 2 W	1	16.00	16.00
Prototype board and wiring	1	5.00	5.00
Raspberry Pi Zero expansion / battery module	1	40.00	40.00
Total			95.90

4.2 Calibration results

Across ten pendulum repetitions, the instrumented accelerometer showed an average error of approximately 3.5% relative to the theoretical pendulum calculation and approximately 94% similarity to the commercial BWT61CL accelerometer. Representative time-series traces from the instrumented and reference accelerometers are shown in Figure 5.

Table 4 summarizes the static load-cell tests reported in the project. The 10lb test is listed using the standard conversion of $10\text{ lb} \approx 4536\text{ g}$; the original project narrative also reported an aggregate 14% mean error for load-cell calibration. This discrepancy should be reconciled with

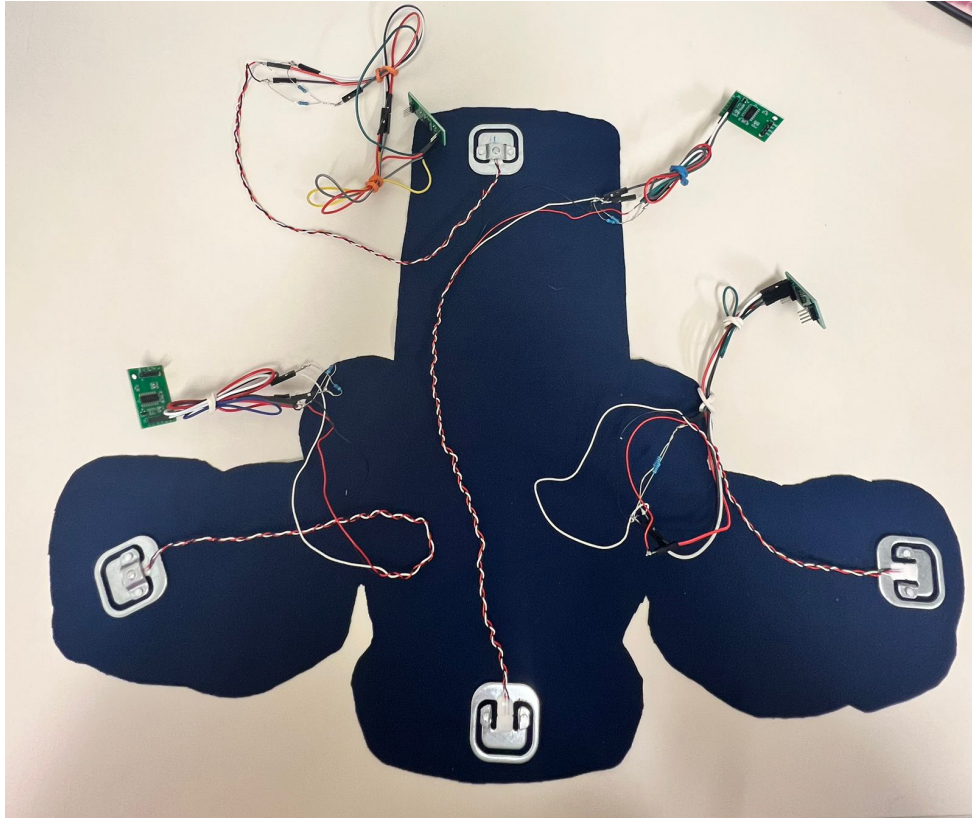


Figure 4: Assembled fabric insert with four load cells and wiring. The design is intended to sit beneath the internal helmet structures and padding.

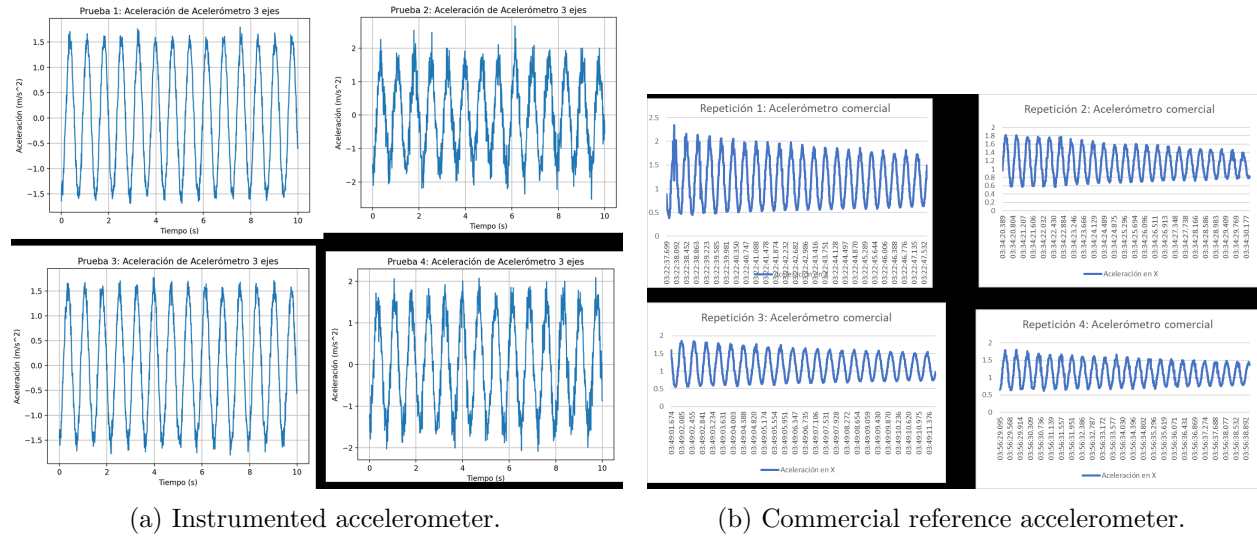


Figure 5: Representative acceleration traces collected during pendulum validation. The repeated sinusoidal responses indicate consistent acquisition during controlled motion.

raw data before journal submission, especially because static calibration does not capture dynamic helmet impacts.

Table 4: Static load-cell calibration summary from known-mass tests.

Nominal test mass	Expected mass (g)	Reported mean (g)	Relative error (%)
Smartphone reference	190	192.0	1.1
2.5 lb disk	1133	1134.462	0.1
5 lb disk	2267	2263.575	-0.2
10 lb disk	4536	4534.693	0.0

A measurement delay of approximately 12 s was observed during the load-cell calibration workflow. That delay is not adequate for real-time sideline alerting and should be reduced through optimized data acquisition, interrupt handling, and buffering before field use.

4.3 Technology readiness

The prototype reached a proof-of-concept level consistent with Technology Readiness Level 3: analytical and experimental critical-function validation in a laboratory setting. The system demonstrated sensing, thresholding, and display logic, but was not evaluated in live football practice, controlled helmet-drop testing, or athlete-worn field trials.

5 Discussion

The prototype demonstrates that a low-cost helmet-integrated impact-monitoring system can be built with accessible electronics and open programming tools. The combined accelerometer and load-cell approach provides two complementary measures: global head acceleration for event detection and local load response for approximate impact-region identification. This architecture is simpler and less expensive than many commercial telemetry systems, making it attractive for early-stage research, engineering education, and resource-constrained teams.

The most important limitation is that the system was validated only in benchtop conditions. Sport-related concussion cannot be inferred from a single acceleration value, and linear acceleration alone omits rotational kinematics, impact duration, athlete-specific biomechanics, previous concussion history, and clinical symptoms. The current prototype therefore should be considered a screening adjunct. When an alert occurs, the safer workflow is removal from play and evaluation by qualified medical personnel using a validated assessment pathway [1, 2].

The load-cell subsystem also requires additional development. The static mass tests suggest that the sensors can be calibrated under controlled conditions, but helmet impacts are transient, multi-directional, and affected by shell deformation, padding, sensor mounting, and wiring strain. Dynamic calibration should use a repeatable impact rig or helmet drop tower with synchronized reference instrumentation. Future work should also characterize cross-talk among load cells, sample latency, saturation, hysteresis, and the influence of helmet fit.

A second limitation is ergonomics and safety. Any device placed inside a helmet can alter fit, comfort, and energy-management behavior. Before athlete use, the system should undergo fit assessment, enclosure redesign, electrical insulation, moisture protection, thermal testing, and review against helmet certification constraints. The prototype should not be used in competition until the added components are shown not to compromise helmet performance.

The ethical considerations are similarly central. Impact records and survey responses may contain health-related information, so data access, storage duration, consent, and anonymization must be specified before any human-subject data collection. A future multi-athlete version should include encrypted storage, participant identifiers separated from sensor data, role-based access, and clear procedures for data deletion.

6 Future work

Future development should prioritize five tasks. First, a dynamic impact validation protocol should compare the prototype against a calibrated reference system across frontal, rear, and lateral impact locations. Second, rotational kinematics should be added through a gyroscope or multi-sensor algorithm to better approximate head motion. Third, the acquisition code should be optimized to reduce load-cell delay and enable reliable real-time alerting. Fourth, the user interface should support multiple athletes, session-level reports, and clinical follow-up notes. Fifth, the device should be evaluated in a controlled field study with ethics approval, medical oversight, and video synchronization to verify whether detected events correspond to observed impacts.

7 Conclusion

This article presents a low-cost helmet-integrated prototype for measuring head impacts in American football. The system combines an H3LIS200DL accelerometer, four load cells, HX711 modules, and a Raspberry Pi Zero 2 W to detect events above 14.4 g, estimate impact region, and display sideline alerts. Preliminary benchtop testing showed approximately 94% agreement with a commercial accelerometer and demonstrated feasible static load-cell calibration. The current prototype is suitable for proof-of-concept research but not for clinical diagnosis or unsupervised athlete use. With dynamic validation, improved sensor timing, field-ready packaging, and integration into medical workflows, the approach could support more accessible head-impact monitoring in youth and collegiate football programs.

Declarations

Ethics and consent. The source project included an anonymous survey of coaching staff and an informed-consent process. Any future athlete-worn or health-data study should obtain institutional ethics approval before data collection.

Clinical disclaimer. The prototype is a screening and research device only. It does not diagnose concussion and must not determine return-to-play status.

Data availability. Raw sensor logs and videos were referenced in the source project repository; they should be archived in a stable, access-controlled repository before journal submission.

Funding. No external funding information was provided in the source report.

Conflicts of interest. The authors should declare any financial or commercial relationships before submission.

References

- [1] Jon S. Patricios, Kathryn J. Schneider, Jiri Dvorak, Osman H. Ahmed, Cheri Blauwet, Robert C. Cantu, Gavin A. Davis, Ruben J. Echemendia, Michael Makdissi, Mike McNamee, Margot

- Putukian, et al. Consensus statement on concussion in sport: the 6th international conference on concussion in sport—amsterdam, october 2022. *British Journal of Sports Medicine*, 57(11): 695–711, 2023. doi: 10.1136/bjsports-2023-106898.
- [2] Ruben J. Echemendia, Bruce L. Brett, Steven P. Broglio, Gavin A. Davis, Christopher C. Giza, Michael McCrea, Paul McCrory, William P. Meehan, Jon S. Patricios, Margot Putukian, et al. Sport concussion assessment tool 6 (scat6). *British Journal of Sports Medicine*, 57(11):622–631, 2023. doi: 10.1136/bjsports-2023-106849.
 - [3] Kevin M. Guskiewicz, Stephen W. Marshall, Julian Bailes, Michael McCrea, Robert C. Cantu, Christopher Randolph, and Barry D. Jordan. Association between recurrent concussion and late-life cognitive impairment in retired professional football players. *Neurosurgery*, 57(4): 719–726, 2005. doi: 10.1227/01.NEU.0000175725.75780.DD.
 - [4] Kevin M. Guskiewicz, Stephen W. Marshall, Julian Bailes, Michael McCrea, Henry P. Harding, Abigail Matthews, Jason R. Mihalik, and Robert C. Cantu. Recurrent concussion and risk of depression in retired professional football players. *Medicine and Science in Sports and Exercise*, 39(6):903–909, 2007. doi: 10.1249/mss.0b013e3180383da5.
 - [5] Jesse Mez, Daniel H. Daneshvar, Patrick T. Kiernan, Bobak Abdolmohammadi, Victor E. Alvarez, Bertrand R. Huber, Michael L. Alosco, Tessa M. Solomon, Christopher J. Nowinski, Lauren McHale, et al. Clinicopathological evaluation of chronic traumatic encephalopathy in players of american football. *JAMA*, 318(4):360–370, 2017. doi: 10.1001/jama.2017.8334.
 - [6] Kevin M. Guskiewicz, Jason P. Mihalik, Vinay Shankar, Stephen W. Marshall, David H. Crowell, Susan M. Oliaro, Michael F. Ciocca, and David N. Hooker. Measurement of head impacts in collegiate football players: relationship between head impact biomechanics and acute clinical outcome after concussion. *Neurosurgery*, 61(6):1244–1253, 2007. doi: 10.1227/01.neu.0000306103.68635.1a.
 - [7] Steven P. Broglio, Jacob J. Sosnoff, Seungmo Shin, Xin He, Carlos Alcaraz, and James Zimmerman. Head impacts during high school football: a biomechanical assessment. *Journal of Athletic Training*, 44(4):342–349, 2009. doi: 10.4085/1062-6050-44.4.342.
 - [8] Kody R. Campbell. Quantifying and comparing the head impact biomechanics of different player positions for canadian university football. Master’s thesis, University of Ottawa, 2014.
 - [9] Elliot J. Pellman, David C. Viano, Andrew M. Tucker, Ira R. Casson, and Joseph F. Waeck-erle. Concussion in professional football: location and direction of helmet impacts—part 2. *Neurosurgery*, 53(6):1328–1341, 2003. doi: 10.1227/01.NEU.0000093499.20604.21.
 - [10] David C. Viano, Ira R. Casson, and Elliot J. Pellman. Concussion in professional football: biomechanics of the struck player—part 14. *Neurosurgery*, 61(2):313–328, 2007. doi: 10.1227/01.NEU.0000279969.02685.D0.
 - [11] David C. Viano, Ira R. Casson, and Elliot J. Pellman. Concussion in professional football: biomechanics of the striking player—part 8. *Neurosurgery*, 56(2):266–280, 2005. doi: 10.1227/01.NEU.0000147865.75357.46.
 - [12] Gamil Rodriguez Rodriguez et al. Sistema para la medicion de deformacion usando galgas extensiométricas metálicas. *Ingeniería Electronica, Automática y Comunicaciones*, 37(1):80–93, 2016.

- [13] DFRobot. H3LIS200DL Triple Axis Accelerometer SKU:SEN0408. https://wiki.dfrobot.com/H3LIS200DL_Triple_Axis_Accelerometer_SKU_SEN0408, 2023. Technical documentation.
- [14] Raspberry Pi Ltd. Raspberry Pi Zero 2 W product brief. <https://www.raspberrypi.com/products/raspberry-pi-zero-2-w/>, 2021. Product documentation.